Track 47: Electric Vehicle Technology

June 2026

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Tier 3 — Specialized | Departments: EEE · Mechanical · Mechatronics · ECE

> ***Skill path:*** *EV Architecture → Battery Systems → BMS → Motor Drives → Power Electronics → Charging → MATLAB Simulation*

Related: Track 39 — Power Systems · Track 38 — MATLAB/Simulink · Track 25 — PLC

Track Overview

Who this is for: EEE, Mechanical, Mechatronics, and ECE students targeting EV industry roles (OEM, charging infra, battery tech).

Prerequisites: Basic electrical machines and circuits; Track 39 M1 helpful.

Tools: MATLAB/Simulink, battery test data, motor drive simulators, EV schematic tools, lab kits (optional).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	EV awareness	EV subsystem block diagram + sim
1 Month	80–100	Junior EV technician/analyst	Battery + motor drive study
3 Months	250–300	EV intern ready	Integrated EV powertrain capstone
6 Months	500–600	EV engineer entry	Full EV system design portfolio

2-Week Crash Course

Days	Topics	Deliverable
D1–2	EV architecture, BEV/PHEV/HEV comparison	EV technology comparison report
D3–4	Li-ion batteries, BMS functions, safety	Battery spec analysis
D5–6	BLDC/PMSM motors, inverters, VFD	Motor drive simulation
D7–8	Charging: AC Level 1/2, DC fast charging	Charging infrastructure brief
D9–10	Capstone	**EV powertrain Simulink model**

1-Month Foundation

Week	Topics	Project
W1	Battery chemistry, SOC/SOH, thermal management	BMS logic diagram
W2	Motor control: FOC, regenerative braking	Drive cycle simulation
W3	Power electronics: DC-DC, onboard charger	Power flow analysis
W4	EV standards, safety (ISO 26262 intro), capstone	**EV system design document**

3-Month Job Ready

Powertrain integration + range estimation + charging station design intro + capstone with MATLAB/Simulink full vehicle model.

6-Month Professional

Add: battery pack sizing project, CAD integration (Track 52), charging network case study, industry visit report.

Assessment Rubric

Weekly Quiz 20% | Labs/simulation 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Cross-department Tier 3 course. EEE focus: power electronics + motors. Mechanical focus: thermal + chassis integration. Mechatronics: control systems.